Windows Logging for SOC — Incident Investigation Report

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Investigation Type : Windows Log Analysis / Threat Hunting / DFIR

Platform : Windows

Lab : TryHackMe — Windows Logging for SOC

Date : September 2026

Executive Summary

This report documents the investigation of suspicious activity identified on a Windows workstation during the Windows Logging for SOC laboratory. The objective of the investigation was to analyze Windows Security logs, Sysmon telemetry, and PowerShell artifacts to identify indicators of compromise, reconstruct attacker behavior, and determine the overall security impact.

The investigation identified evidence of:

- Repeated failed authentication attempts
- Successful malicious RDP authentication
- Unauthorized account creation
- Privilege escalation activity
- Suspicious process execution
- Malware download activity
- Persistence mechanisms
- Command-and-control communication
- Malicious PowerShell usage

Correlation across multiple Windows log sources enabled reconstruction of the attacker timeline and identification of malicious behavior consistent with a full intrusion lifecycle

1. Investigation Objectives

The primary objectives of the investigation were:

- Detect unauthorized authentication attempts
- Investigate successful logon events
- Identify suspicious process execution
- Analyze Sysmon telemetry
- Detect privilege escalation behavior
- Investigate persistence mechanisms
- Reconstruct attacker timeline
- Identify indicators of compromise (IOCs)
- Recommend mitigation and detection improvements

2. Log Sources Investigated

2.1 Windows Security Logs

Authentication and account activity were analyzed using:

Event Viewer → Windows Logs → Security

Purpose

1. Failed logon detection
2. Successful authentication analysis
3. RDP login monitoring
4. User management investigation
5. Privileged group modifications

Key Event IDs

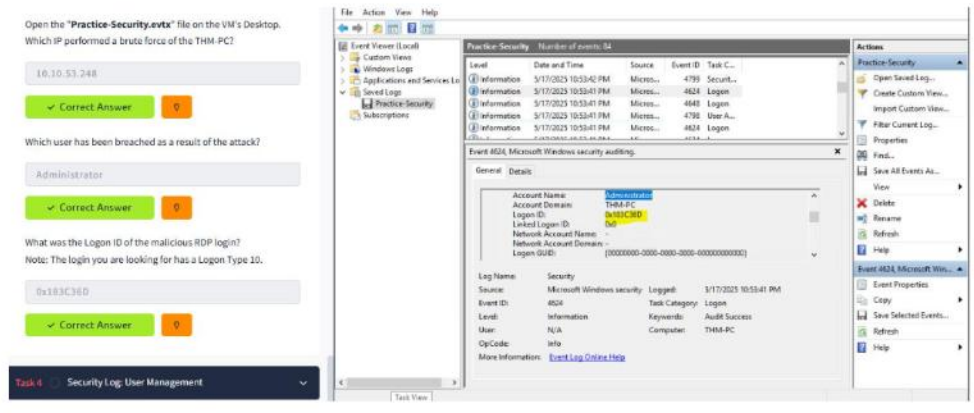
4624 — Successful Logon

4625 — Failed Logon

4720 — User Account Created

4732 — User Added to Privileged Group

Screenshot Placement



2.2 Sysmon Operational Logs

Sysmon telemetry was analyzed using:

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational

Purpose

1. Process creation monitoring
2. Network connection tracking
3. File creation monitoring
4. DNS request analysis
5. Malware execution detection

Key Event IDs

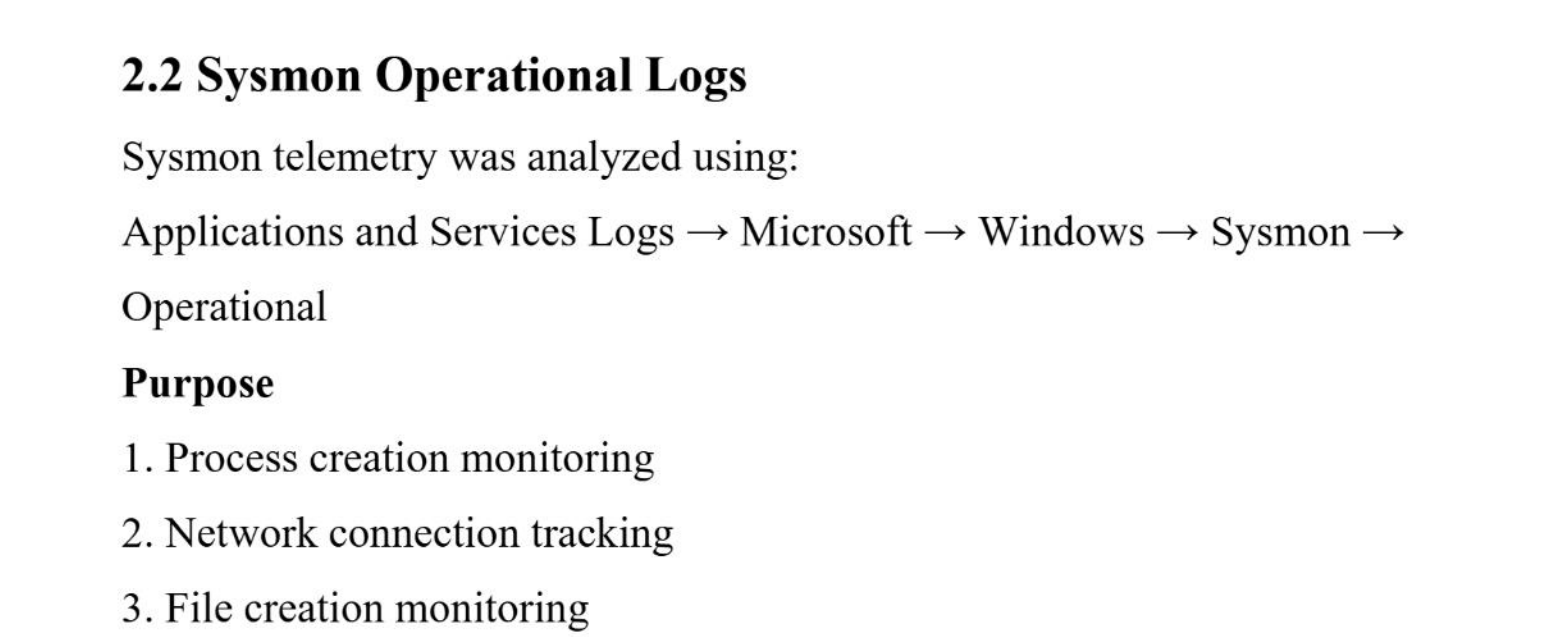
Event ID 1 — Process Creation

Event ID 3 — Network Connection

Event ID 11 — File Creation

Event ID 22 — DNS Query

Screenshot Placement



2.3 PowerShell History Analysis

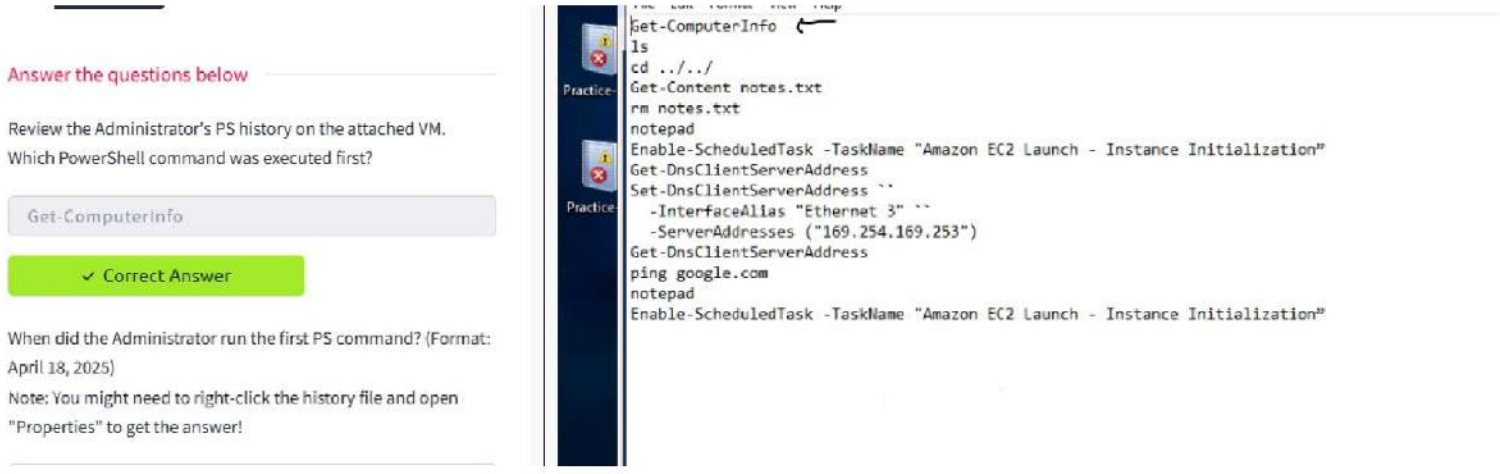
PowerShell activity was investigated using:

`C:\Users\<USER>\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadline\ConsoleHost_history.txt`

Purpose

1. PowerShell command auditing
2. Malware download detection
3. Administrative command analysis
4. User activity investigation

Screenshot Placement



3. Investigation Methodology

The investigation followed a standard SOC triage and DFIR workflow:

1. Review Windows Security logs
2. Identify failed authentication attempts
3. Investigate successful RDP logons
4. Correlate suspicious Logon IDs
5. Analyze account management activity
6. Review Sysmon process execution logs
7. Investigate network communication events
8. Analyze PowerShell history artifacts
9. Reconstruct attacker timeline
10. Document findings and evidence

4. Authentication Analysis

4.1 Failed Authentication Attempts

The following filtering activity was performed:

Event ID: 4625

Findings

Multiple failed login attempts were identified targeting the Windows host through Remote Desktop Protocol (RDP). The attack originated from a suspicious external IP address.

Analysis

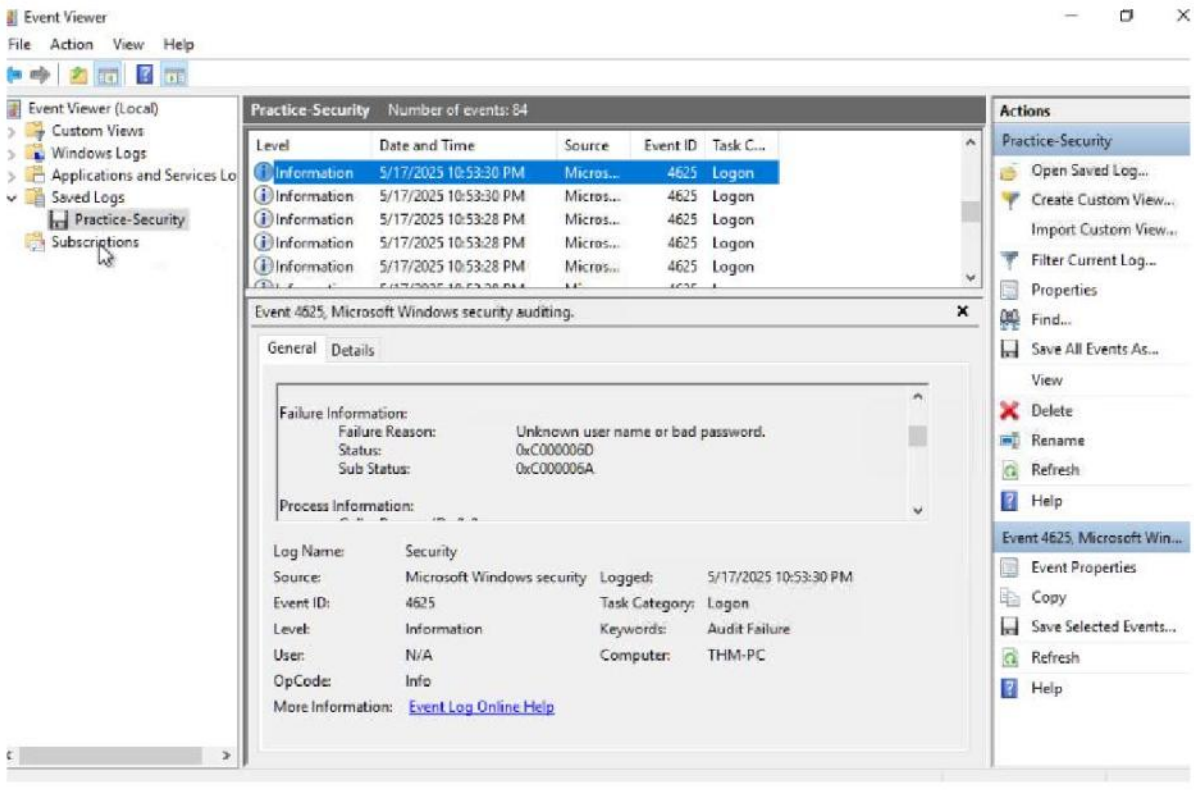
Observed behavior is consistent with:

- RDP brute-force attacks
- Password spraying
- Credential stuffing attempts

Security Impact

Repeated failed authentication attempts indicate active adversary probing and unauthorized access attempts against the Windows workstation.

Screenshot Placement



4.2 Successful Malicious RDP Authentication

The following filter was applied:

Event ID: 4624

Findings

A successful RDP login was identified shortly after the failed authentication attempts. Logon Type 10 confirmed remote interactive access through RDP.

Analysis

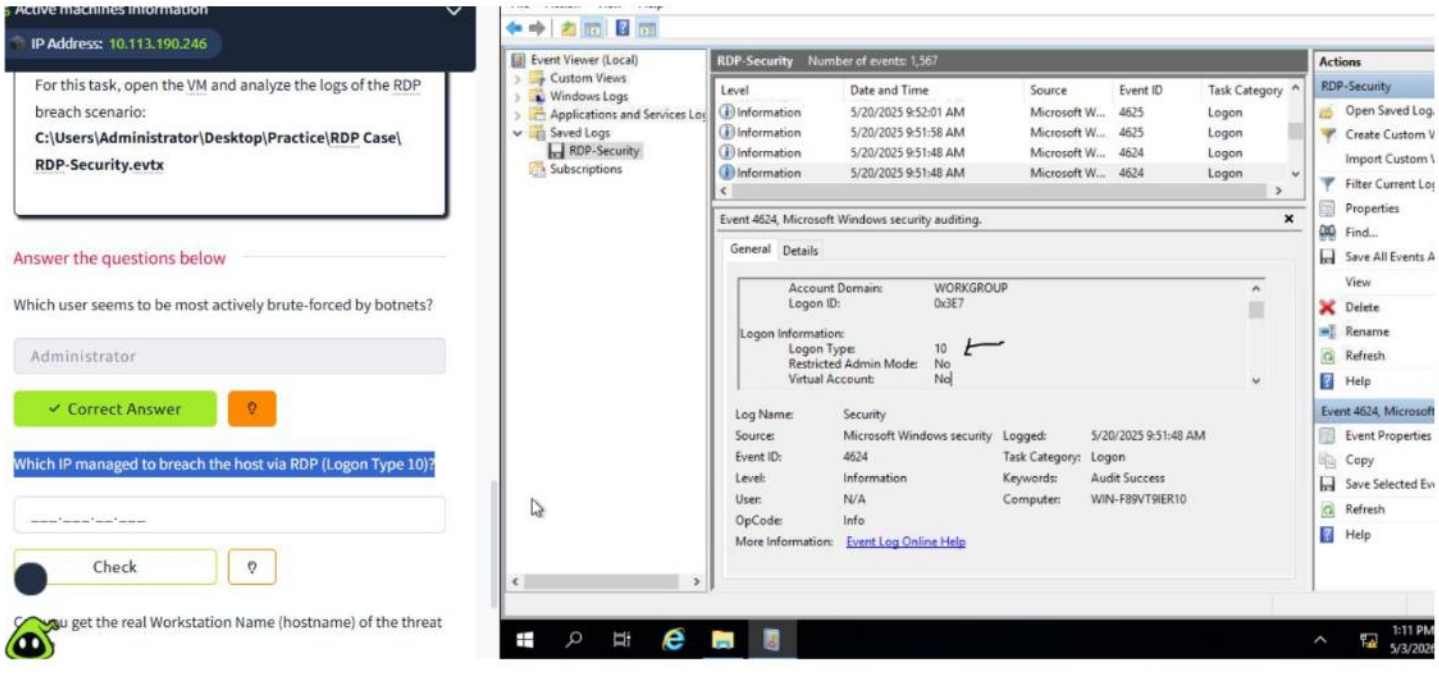
The authentication sequence strongly suggests:

- Credential compromise
- Successful attacker access
- Initial foothold establishment

Security Impact

Successful malicious RDP authentication significantly increases the probability of full host compromise.

Screenshot Placement



4.3 Logon ID Correlation

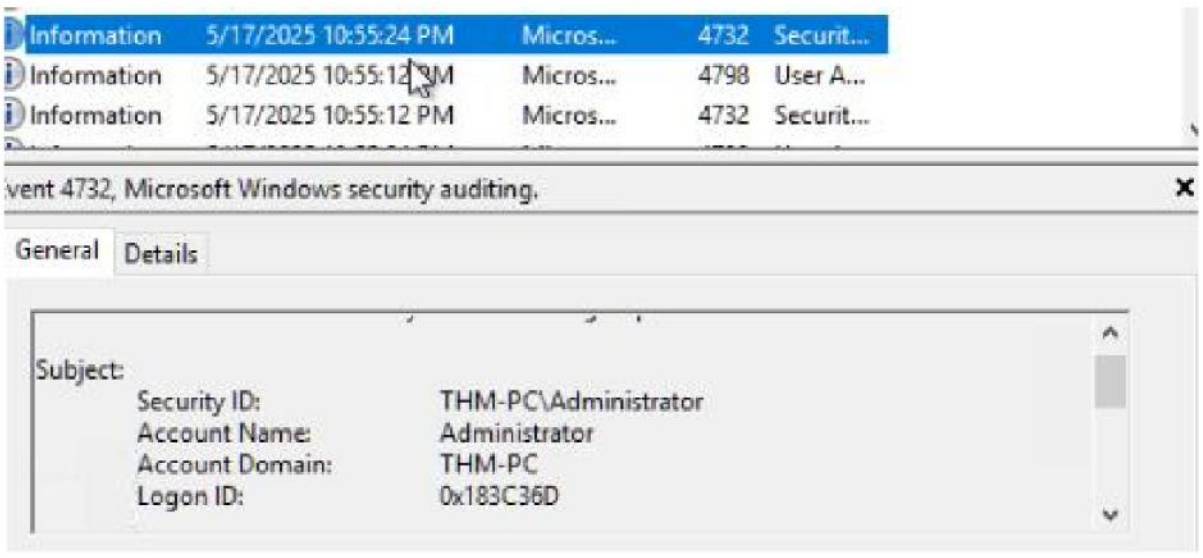
Findings

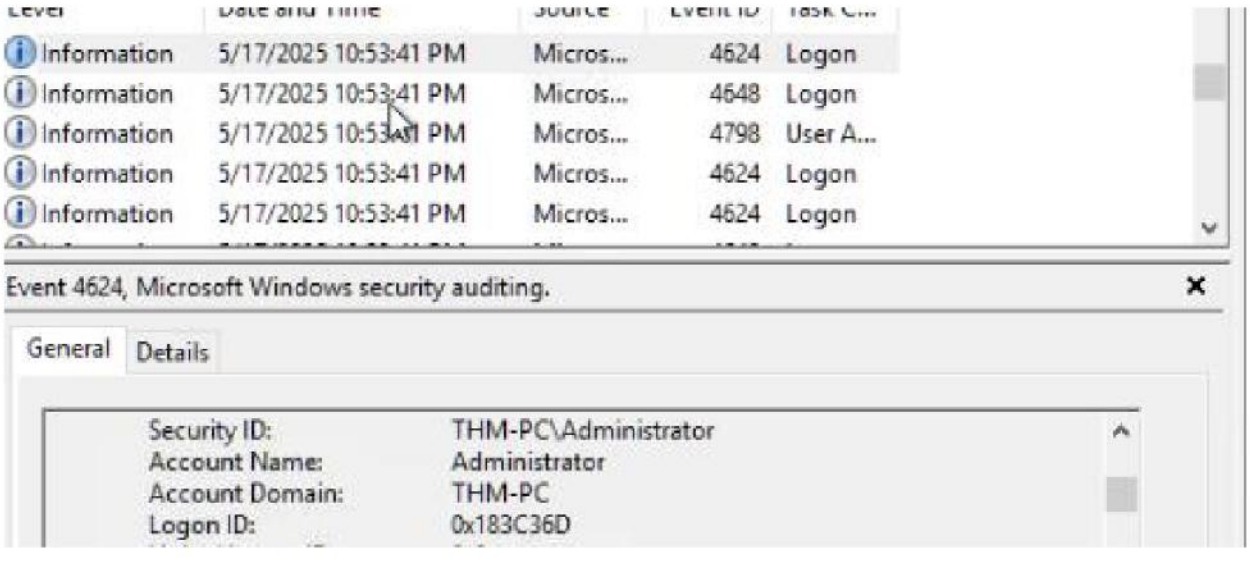
The malicious session was tracked using the Logon ID associated with the attacker's RDP session.

Analysis

Correlation of Logon IDs enabled tracking of:

- User management activity
- Process execution
- Persistence establishment
- Privilege escalation operations





5. User Management Investigation

5.1 Unauthorized Account Creation

The following Event ID was investigated:

4720 — User Account Created

Findings

A suspicious user account was created shortly after the successful RDP login.

Analysis

1. Observed behavior indicates:

2. Persistence establishment

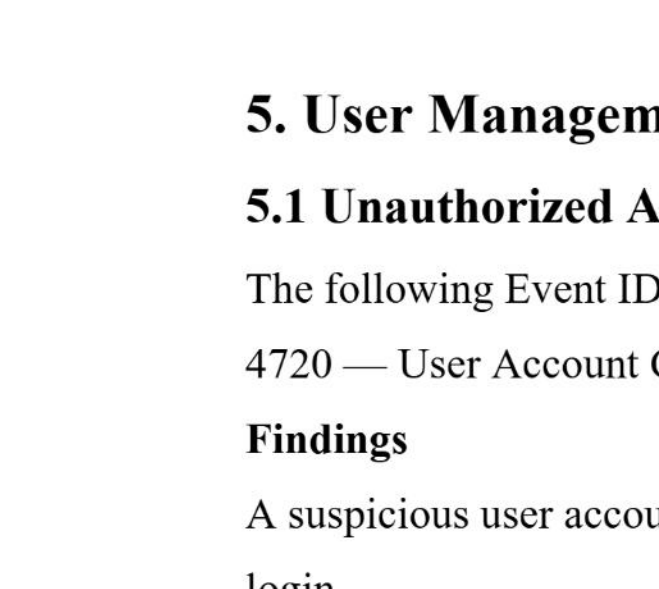
3. Backdoor account creation

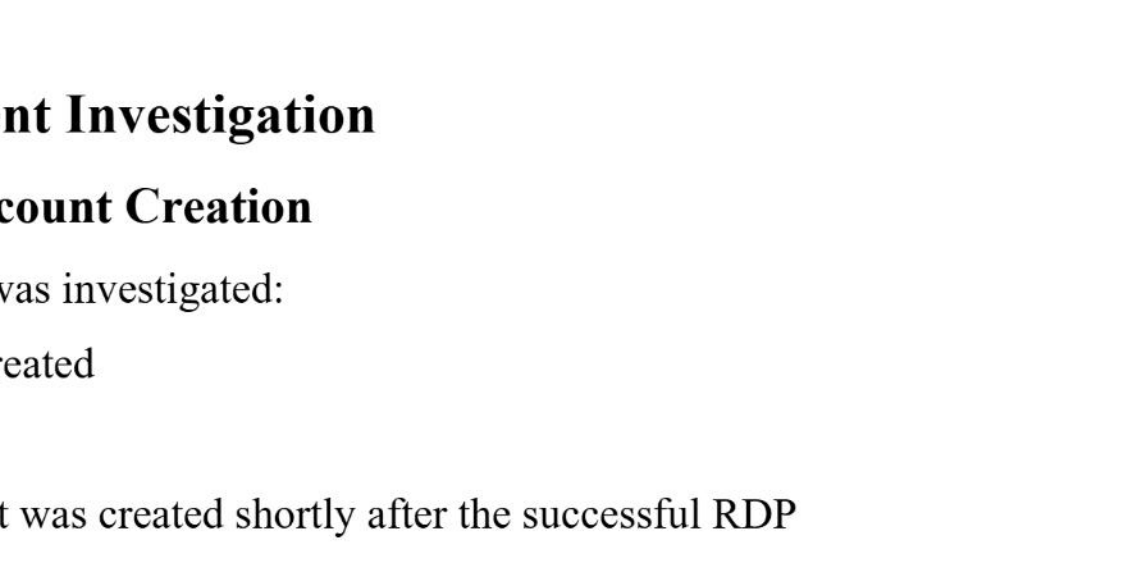
4. Long-term unauthorized access preparation

Security Impact

Unauthorized account creation allows attackers to maintain persistent access even after password resets.

Screenshot Placement





5.2 Privileged Group Membership Modification

The following Event ID was analyzed:

4732 — Member Added to Security-Enabled Local Group

Findings

The newly created account was added to privileged local groups.

Analysis

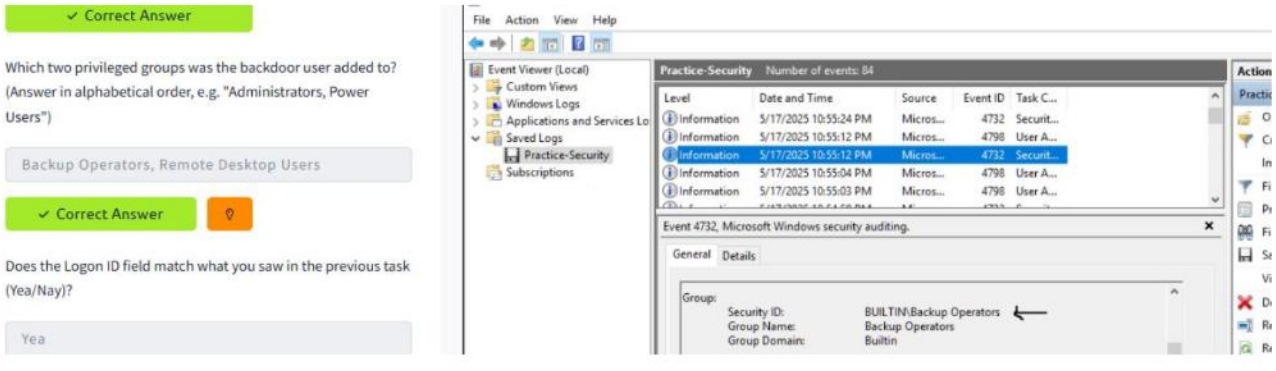
This activity indicates:

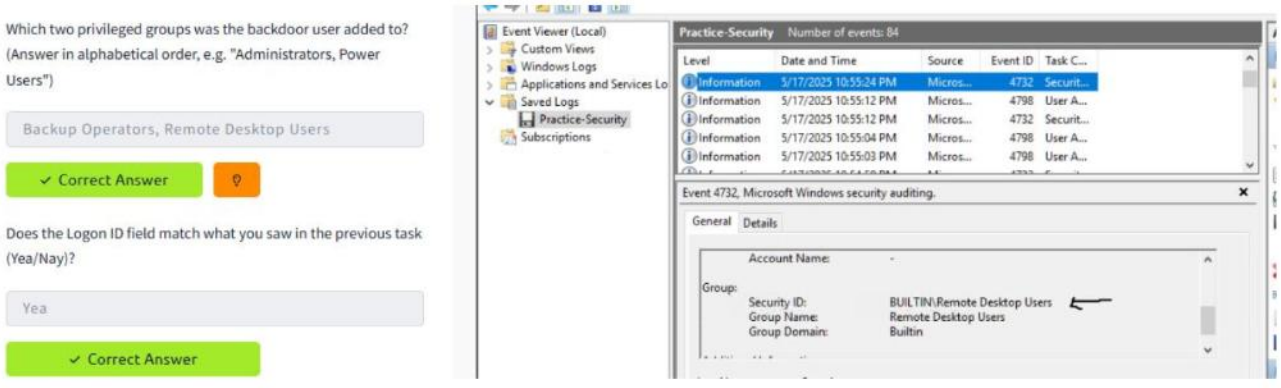
- Privilege escalation
- Administrative rights acquisition
- Persistence hardening

Security Impact

Attackers with administrative group membership gain full control over the compromised system.

Screenshot Placement





6. Sysmon Investigation

6.1 Process Creation Analysis

The following Sysmon event was analyzed:

Sysmon Event ID 1 — Process Creation

Findings

Suspicious process execution activity involving browser downloads and malware execution was identified.

Analysis

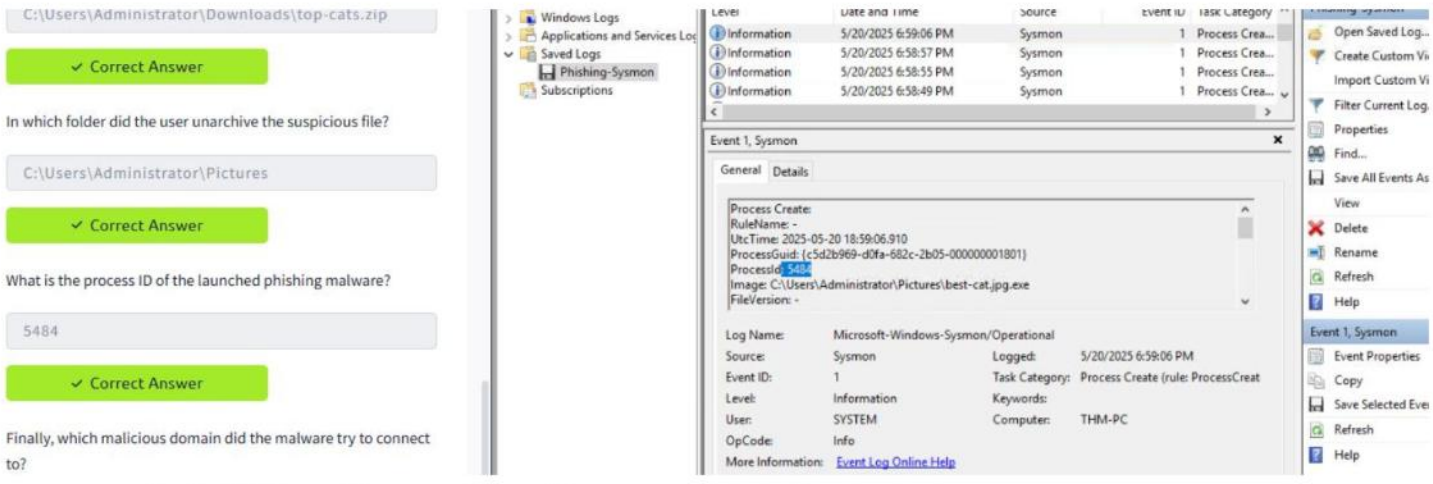
Observed behavior suggests:

- Malware download activity
- Payload execution
- User-driven compromise chain

Security Impact

Malicious process execution may allow attackers to deploy persistence mechanisms and remote access malware.

Screenshot Placement



6.2 File Creation Investigation

The following Sysmon event was reviewed:

Sysmon Event ID 11

Findings

File creation activity was observed during Sysmon analysis.

Analysis

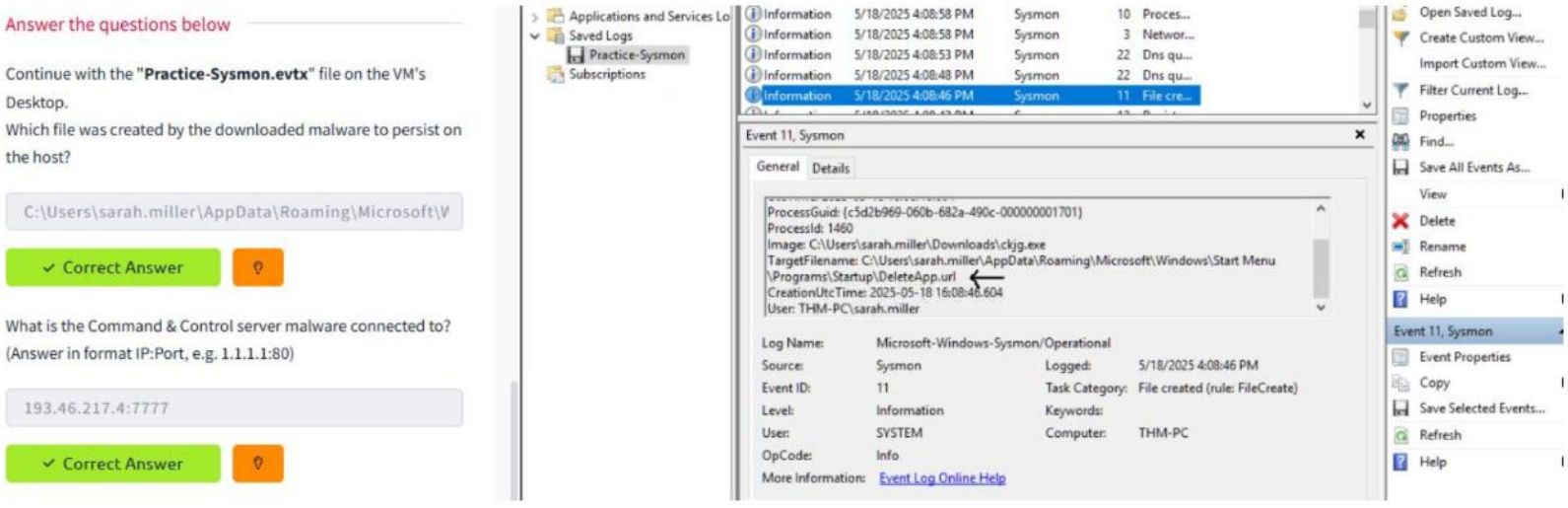
The event provided visibility into:

- File paths
- Creation timestamps
- Associated processes

Security Impact

File creation monitoring assists with identifying newly introduced files and suspicious system activity.

Screenshot Placement



6.3 Network Communication Analysis

The following event was analyzed:

Sysmon Event ID 3 — Network Connection

Findings

Outbound network connections to suspicious infrastructure were identified.

Analysis

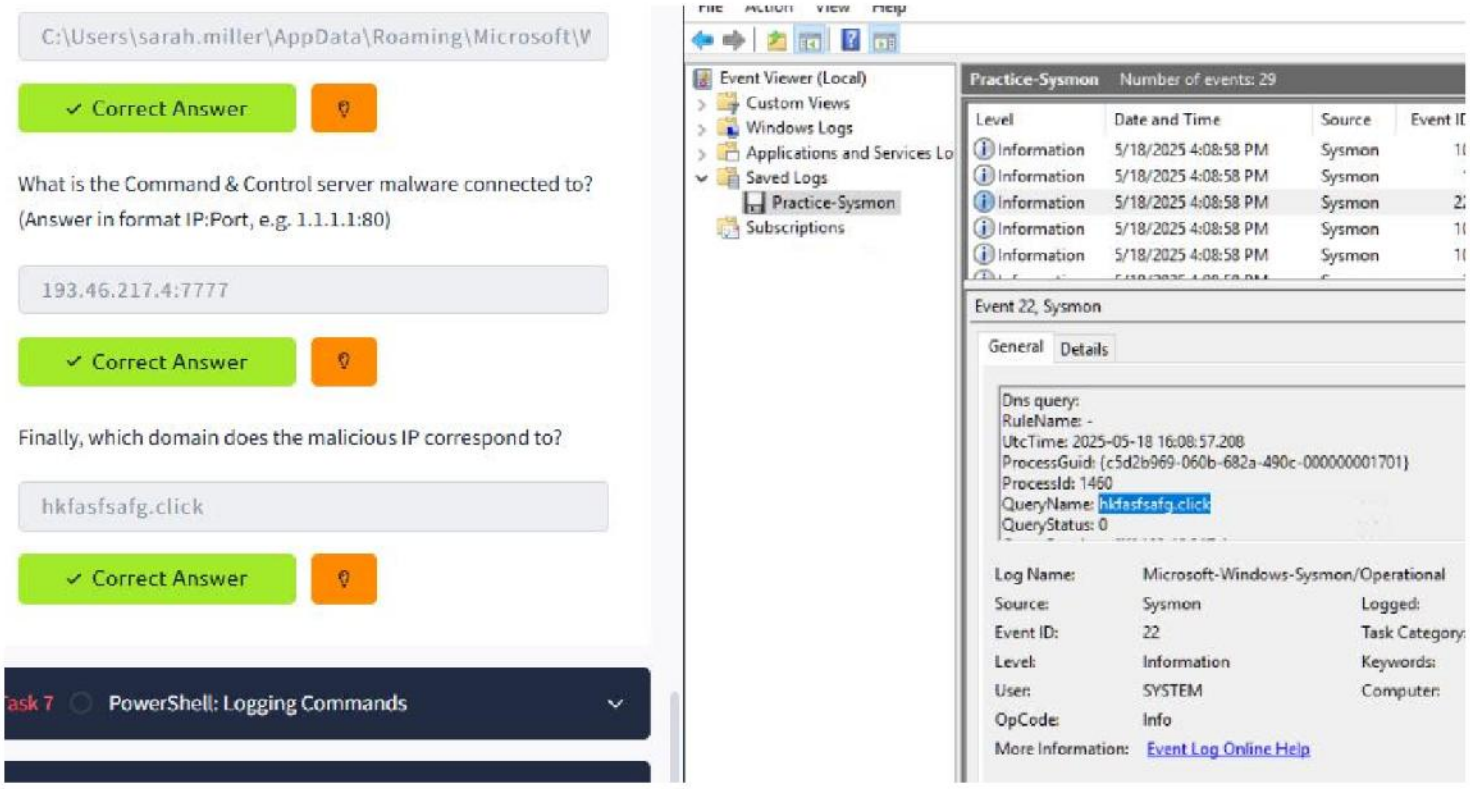
Observed behavior indicates:

- Command-and-control communication
- Malware beaconing
- External attacker interaction

Security Impact

Command-and-control traffic confirms active malware communication with attacker infrastructure.

Screenshot Placement



7. PowerShell Investigation

7.1 PowerShell History Analysis

The following artifact was investigated:

ConsoleHost_history.txt

Findings

PowerShell history revealed administrative and reconnaissance-related commands executed on the compromised host.

Observed Commands

Get-ComputerInfo

Get-LocalUser

Get-LocalGroup

Invoke-WebRequest

Analysis

The commands indicate:

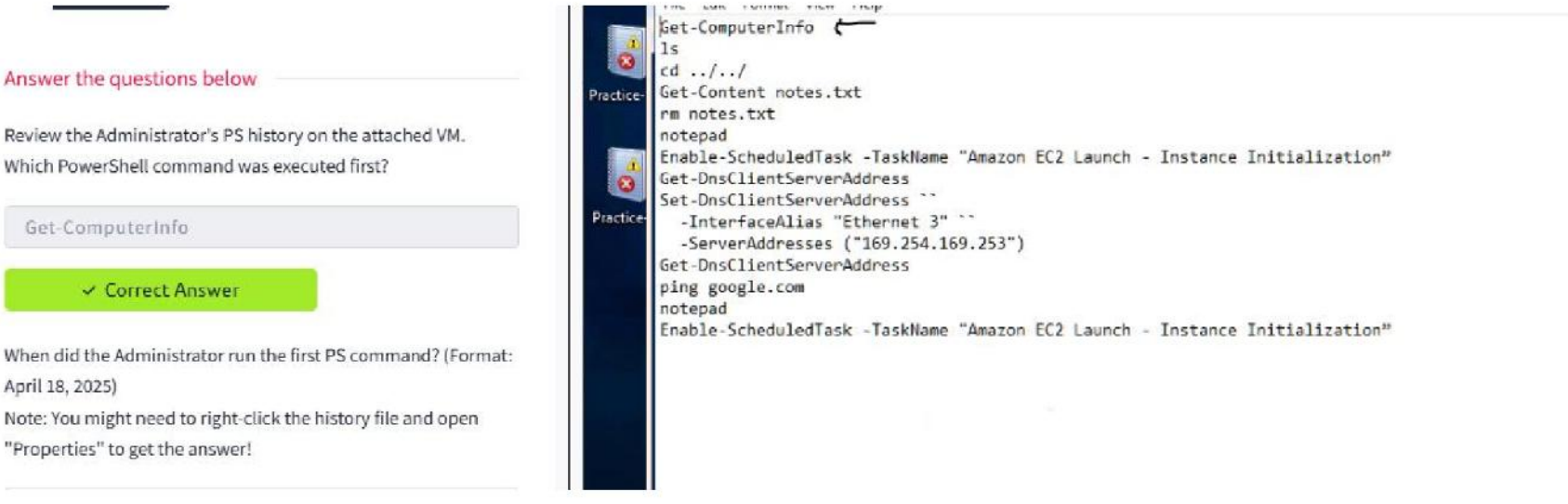
- System reconnaissance
- User enumeration
- Malware download activity

- Administrative discovery operations

Security Impact

PowerShell abuse remains one of the most common attacker techniques in Windows intrusions.

Screenshot Placement

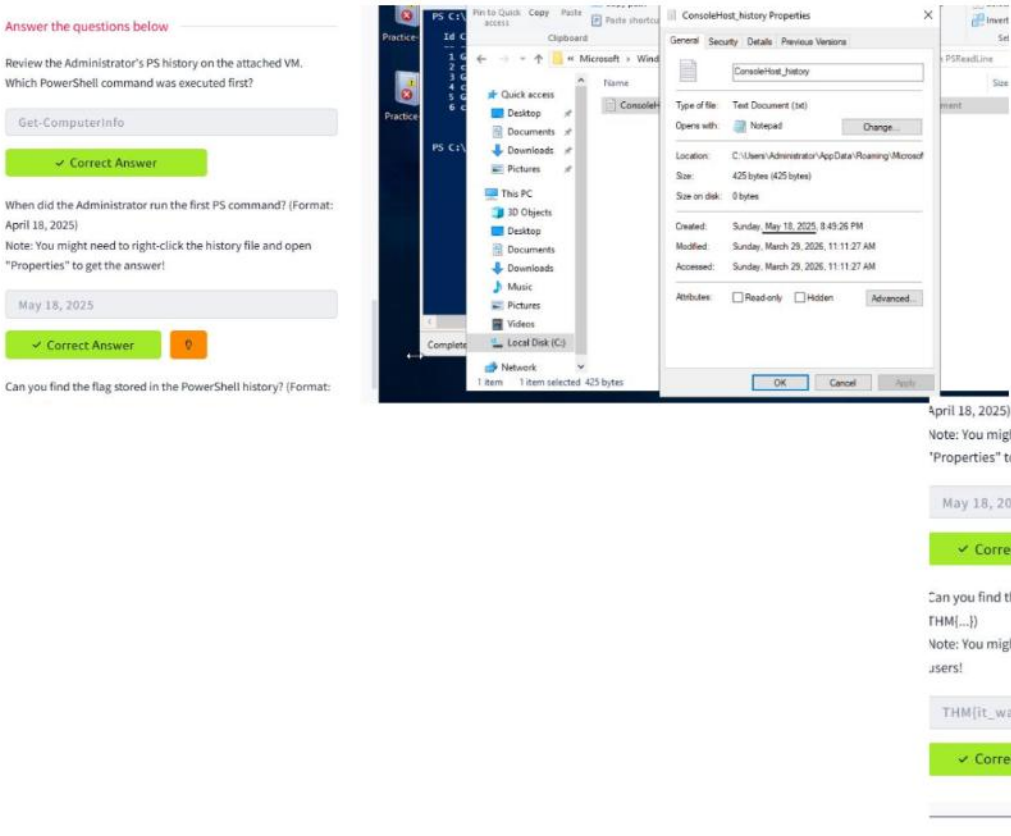


7.2 PowerShell Artifact Timestamp Analysis

Findings

The PowerShell history file creation timestamp confirmed the approximate time of initial PowerShell activity.

Screenshot Placement



9. Indicators of Compromise (IOCs)

Network Indicators

Suspicious External IP	Source of brute-force authentication activity
Malicious Domain	Command-and-control infrastructure

Process Indicators

powershell.exe	PowerShell execution
browser.exe	Malware download activity
suspicious executable	Malware payload

10. Behavioral Indicators

- Repeated RDP authentication failures
- Successful remote login after brute-force activity
- Unauthorized account creation
- Privileged group modification
- Suspicious process execution
- Malware download behavior
- PowerShell reconnaissance activity
- Command-and-control communication

12. Risk Assessment

Category	Severity
Unauthorized Access	High
Privilege Escalation	High
Persistence Risk	High
Malware Execution	High
Data Exposure	Medium
Command-and-Control Activity	High

13. Recommendations

- Immediate Actions
- Disable compromised accounts
- Reset exposed credentials
- Remove unauthorized users
- Block malicious IP addresses
- Isolate affected systems
- Conduct malware eradication procedures

14. Long-Term Improvements

- Enable Sysmon enterprise-wide
- Centralize logs into SIEM
- Configure MFA for RDP access
- Harden Windows authentication policies
- Monitor privileged group modifications
- Enable PowerShell logging controls
- Implement endpoint detection and response (EDR)

15. Detection Opportunities

Brute Force Detection

Monitor for:

Multiple Event ID 4625 failures from same source IP

Privileged Group Monitoring

Monitor for:

Event ID 4732 modifications involving Administrators group

PowerShell Abuse Detection

Monitor for:

- Invoke-WebRequest usage
- Encoded PowerShell commands
- Suspicious PowerShell history artifacts

16. Conclusion

The investigation identified multiple indicators of malicious activity on the Windows host.

Security log analysis revealed repeated brute-force authentication attempts followed by successful RDP compromise. Subsequent account creation and privileged group modification activity confirmed persistence establishment and privilege escalation.

Sysmon telemetry identified malware download activity, suspicious process execution, persistence mechanisms, and outbound command-and-control communication.

PowerShell history artifacts further confirmed attacker reconnaissance and operational activity on the host.

Correlation across Windows Security logs, Sysmon telemetry, and PowerShell artifacts enabled complete reconstruction of the attacker lifecycle and demonstrated the effectiveness of Windows logging in SOC investigations.

17. Analyst Notes

This investigation demonstrated practical experience in:

- Windows Event Log analysis
- SOC investigation methodology
- Threat hunting workflows
- Sysmon telemetry analysis
- Authentication monitoring
- PowerShell investigation
- Timeline reconstruction
- Incident reporting